### 5.3.2 General rules for differentiation

**(i) Addition rule**

$$\frac{d}{dx}\left[f(x)+g(x)\right]=\frac{d}{dx}\left[f(x)\right]+\frac{d}{dx}\left[g(x)\right]$$

**(ii) Subtraction rule (or) Difference rule**

$$\frac{d}{dx}\left[f(x)-g(x)\right]=\frac{d}{dx}\left[f(x)\right]-\frac{d}{dx}\left[g(x)\right]$$

**(iii) Product rule**

$$\frac{d}{dx}\left[f(x)\cdot g(x)\right]=\frac{d}{dx}\left[f(x)\right]g(x)+f(x)\frac{d}{dx}\left[g(x)\right]$$

(or) If $u = f(x)$ and $v = g(x)$ then,

$$\frac{d}{dx}(uv)=u\frac{d}{dx}(v)+v\frac{d}{dx}(u)$$

**(iv) Quotient rule**

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right]=\frac{g(x)\frac{d}{dx}\left[f(x)\right]-f(x)\frac{d}{dx}\left[g(x)\right]}{\left[g(x)\right]^2},$$

$g(x)\neq 0$

(or) If $u = f(x)$ and $v = g(x)$ then,

$$\frac{d}{dx}\left(\frac{u}{v}\right)=\frac{v\frac{d}{dx}(u)-u\frac{d}{dx}(v)}{v^2}$$

**(v) Scalar product**

$\frac{d}{dx}\left[cf(x)\right]=c\frac{d}{dx}\left[f(x)\right]$, where $c$ is a constant.

**(vi) Chain rule**

$$\frac{d}{dx}\left[f\left(g(x)\right)\right]=f'\left(g(x)\right)\cdot g'(x)$$

(or) If $y = f(t)$ and $t = g(x)$ then,

$$\frac{dy}{dx}=\frac{dy}{dt}\cdot\frac{dt}{dx}$$

Here we discuss about the differentiation for the explicit functions using standard results and general rules for differentiation.

**Example 5.33**

Differentiate the following functions with respect to $x$.

(i) $x^{\frac{3}{2}}$ (ii) $7e^x$

(iii) $\frac{1-3x}{1+3x}$ (iv) $x^2\sin x$

(v) $\sin^3 x$ (vi) $\sqrt{x^2+x+1}$

***Solution***

(i) $\frac{d}{dx}\left(x^{\frac{3}{2}}\right) = \frac{3}{2}x^{\frac{3}{2}-1}$

$= \frac{3}{2}x^{\frac{1}{2}} = \frac{3}{2}\sqrt{x}$

(ii) $\frac{d}{dx}(7e^x) = 7\frac{d}{dx}(e^x) = 7e^x$

(iii) Differentiating $y = \frac{1-3x}{1+3x}$ with respect to $x$

$$\frac{dy}{dx}=\frac{(1+3x)\frac{d}{dx}(1-3x)-(1-3x)\frac{d}{dx}(1+3x)}{(1+3x)^2}$$

$$=\frac{(1+3x)(-3)-(1-3x)(3)}{(1+3x)^2}$$

$$=\frac{-6}{(1+3x)^2}$$

(iv) Differentiating $y = x^2\sin x$ with respect to $x$

$$\frac{dy}{dx} = x^2\frac{d}{dx}(\sin x)+\sin x\frac{d}{dx}(x^2)$$

$$= x^2\cos x+2x\sin x$$

$$= x(x\cos x+2\sin x)$$

(v) $y = \sin^3 x$ (or) $(\sin x)^3$

Let $u = \sin x$ then $y = u^3$

Differentiating $u = \sin x$ with respect to $x$

We get, $\frac{du}{dx} = \cos x$

Differentiating $y = u^3$ with respect to $u$



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